Project Proposal: Martian-Inspired Tripedal Walking Robot

Abstract

This proposal outlines the design, fabrication, and testing of a low-cost, student-built tripedal walking robot inspired by recent research in adaptive gaits. The system will use servo-driven legs, an onboard microcontroller, and an inertial measurement unit (IMU) to generate and refine walking patterns. The project is targeted at completion within eight weeks and aligns with MTech-level standards for demonstrating applied mechatronics and robotic gait control.

Background & Objective

Recent developments in gait generation highlight the benefits of non-traditional locomotion strategies. A three-legged walking platform offers a unique balance between complexity and mechanical feasibility, making it an ideal research and academic prototype. The primary objectives are: - Design and fabricate a tripedal robot capable of stable walking. - Implement gait generation strategies using oscillators and feedback from IMU. - Evaluate stability and adaptability across different surfaces.

System Overview

- **Structure**: Rigid central base with three articulated legs.
- **Actuation**: 6 high-torque metal gear servos (2 DOF per leg).
- **Control**: Arduino Mega 2560 microcontroller.
- **Sensing**: MPU-6050 IMU for tilt/orientation feedback.
- **Power**: 3S LiPo battery pack.

Bill of Materials & Cost Analysis

Essential Components: - 6 × Metal gear servos (20 kg·cm torque) @ ₹1,367 each = ₹8,202 - Arduino Mega 2560 = ₹2,356 - MPU-6050 IMU = ₹259 - LiPo Battery 3S 2200 mAh = ₹1,640 - 3D printed leg components + aluminium base = ₹2,500 - Wiring, connectors, fasteners = ₹800 - Spare servo, bearings, contingency = ₹1,200

Total Estimated Cost: ₹17,257 (within the ₹15,000–20,000 student-funded range)

Time Schedule (8 Weeks)

- **Week 1–2:** Literature study, CAD modelling, component procurement.
- **Week 3:** 3D print & fabricate structure, assemble base frame.
- **Week 4:** Mount servos, wiring, basic Arduino servo tests.
- **Week 5:** Implement static gaits (tripod stance, step testing).
- **Week 6:** Integrate IMU feedback, tune gait parameters.

- **Week 7:** Trial and error testing on flat and uneven surfaces.
- **Week 8:** Performance evaluation, documentation, and demo.

Expected Outcomes

- Working prototype of a three-legged robot with stable walking.
- Demonstration of adaptive gait control via IMU feedback.
- Proof-of-concept platform for future research in non-traditional locomotion.

Risk Factors & Mitigation

- **Servo burnout** → Buy one spare unit.
- **Power issues** → Use fuses and proper current distribution.
- **Mechanical failures** → 3D print multiple test iterations.

Budget Summary

- Estimated Total: ₹17,257
- Funding: Student-backed
- Scope for ±10% fluctuation due to local procurement

Approval Request

This project represents a feasible, educational, and innovative approach to robotic locomotion. We request faculty approval to proceed with procurement and development.